$$\frac{x}{(-4)(-5)-(3)(-10)} = \frac{y}{(-10)(4)-(-5)(3)} = \frac{1}{(3)(3)-(4)(-4)}$$

$$\frac{x}{(20)-(-30)} = \frac{y}{(-40)-(-15)} = \frac{1}{(9)-(-16)}$$

$$\frac{x}{20+30} = \frac{y}{-40+15} = \frac{1}{9+16}$$

$$\frac{x}{50} = \frac{y}{-25} = \frac{1}{25}$$

Therefore, we get $x = \frac{50}{25}; \quad y = \frac{-25}{25}$

$x = 2; \quad y = -1$

Thus the solution is $x = 2$, $y = -1$.

Verification :

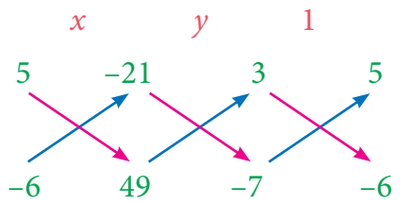
| | | | |
|---|---|---|---|
| $3x-4y$ | = | 10 | ...(1) |
| $3(2)-4(-1)$ | = | 10 | |
| $6+4$ | = | 10 | |
| 10 | = | 10 | True |
| $4x+3y$ | = | 5 | ...(2) |
| $4(2)+3(-1)$ | = | 5 | |
| $8-3$ | = | 5 | |
| 5 | = | 5 | True |

**Example 3.53** Solve by cross multiplication method : $3x+5y=21;\ -7x-6y=-49$

***Solution***

The given system of equations are $3x+5y-21=0;\ -7x-6y+49=0$

Now using the coefficients for cross multiplication, we get,




$$\Rightarrow \frac{x}{(5)(49)-(-6)(-21)} = \frac{y}{(-21)(-7)-(49)(3)} = \frac{1}{(3)(-6)-(-7)(5)}$$

$$\frac{x}{119} = \frac{y}{0} = \frac{1}{17}$$

$$\Rightarrow \quad \frac{x}{119} = \frac{1}{17}, \quad \frac{y}{0} = \frac{1}{17}$$

$$\Rightarrow \quad x = \frac{119}{17}, \quad y = \frac{0}{17}$$

$$\Rightarrow \quad x = 7, \quad y = 0$$

Verification :

| | | | |
|---|---|---|---|
| $3x+5y$ | = | 21 | ...(1) |
| $3(7)+5(0)$ | = | 21 | |
| $21+0$ | = | 21 | |
| 21 | = | 21 | True |
| $-7x-6y$ | = | −49 | ...(2) |
| $-7(7)-6(0)$ | = | −49 | |
| $-49$ | = | −49 | |
| $-49$ | = | −49 | True |

**Note**

Here $\frac{y}{0} = \frac{1}{17}$ is to mean $y = \frac{0}{17}$. Thus, $\frac{y}{0}$ is only a notation and it is not division by zero. It is always true that division by zero is not defined.



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